

- 1 The avocado tree, *Persea americana*, is a cultivated food crop. The fruits of the tree are called avocados.

Figure 1.1 shows a whole avocado and an avocado that has been cut to show the inside.

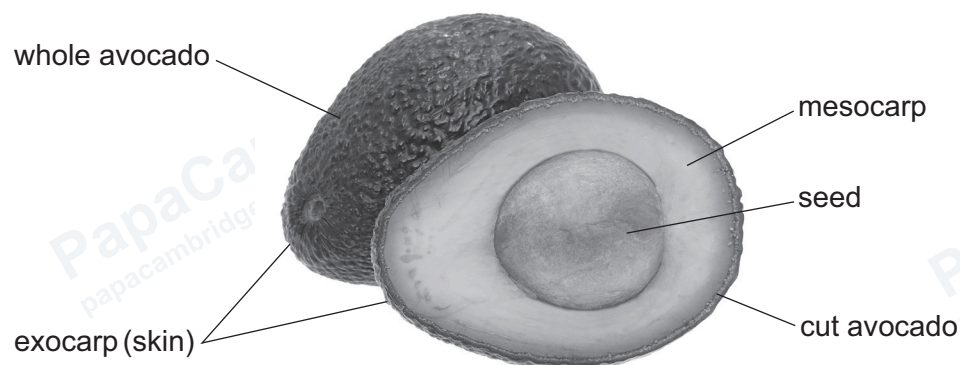


Figure 1.1

Avocados are removed from the tree (harvested) when they are mature (maximum size). Avocados ripen only after harvesting. As avocados ripen, the colour of the skin changes from bright green to dark purple and the mesocarp changes from hard to soft.

Mesocarp cells in the avocado are a source of the enzyme catalase. The quantity of catalase in the mesocarp changes as the avocado ripens.

Catalase catalyses the breakdown of hydrogen peroxide, as shown in Figure 1.2.



Figure 1.2

- (a) A student decided to see the effect of catalase activity in avocados.

The student removed the skin from an avocado and cut up some pieces of the mesocarp. The student placed some of the pieces of mesocarp into a beaker containing hydrogen peroxide solution. Tiny bubbles of oxygen were immediately released from the surface of the mesocarp, as shown in Figure 1.3.

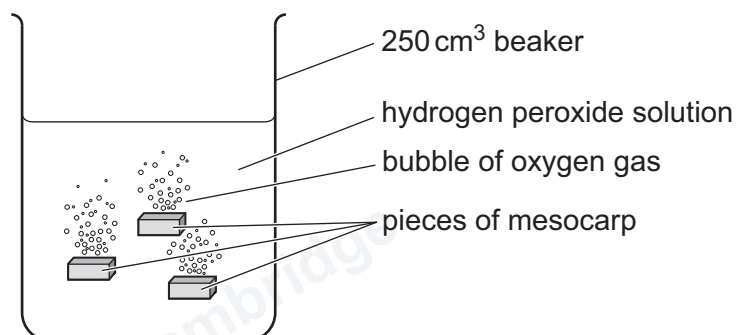


Figure 1.3



Suggest **one** hazard of this method **and** the risk associated with the hazard **and** the precaution that should be taken.

hazard

.....

risk

.....

precaution

.....

[1]



- [1]

- Do **not** include a risk assessment.

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- (c) After food crops such as avocados are harvested, they may be stored in cold conditions (cold storage) to delay ripening.

Some scientists investigated the effect of cold storage for different lengths of time on catalase activity in avocados.

- The scientists selected 100 avocado trees at random from the same field.
- The scientists harvested **one** mature avocado from each tree.
- Each of the 100 avocados had a mass of approximately 210g.
- The catalase activity in 10 of the avocados was measured immediately after harvest.
- The remaining 90 avocados were divided into three groups and stored at 7 °C and 85% humidity for different lengths of time, as shown in Table 1.1.

Table 1.1

group	number of avocados	storage time/days
10d	30	10
20d	30	20
30d	30	30

- (i) Suggest a suitable method the scientists could use to select avocado trees at random.

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 [1]

- (ii) Humidity is the concentration of water vapour in the air. The higher the humidity, the higher the concentration of water vapour in the air.

Suggest a reason why the avocados were stored at 85% humidity.

.....
 [1]

After each storage time, 5 avocados from each group were tested for catalase activity.

The remaining 25 avocados in each group were kept at 20 °C and 80% relative humidity until they were ripe (soft enough to eat).

When they were ripe, 5 avocados from each group were tested for catalase activity.

The results are shown in Table 1.2.

Table 1.2

group	when catalase activity was tested	mean catalase activity /arbitrary units	standard error (SE)	95% confidence interval (95% CI)
0d	immediately after harvesting	571	162	571 ± 324
10d	after 10 days cold storage	615	169	615 ± 338
20d	after 20 days cold storage	455	49	455 ± 98
30d	after 30 days cold storage	550	112	550 ± 224
10dR	after 10 days cold storage plus time to ripen	642	198	642 ± 396
20dR	after 20 days cold storage plus time to ripen	822	206	
30dR	after 30 days cold storage plus time to ripen	736	179	

- (iii) Complete Table 1.2 by calculating the 95% confidence intervals (95% CI) for groups 20dR and 30dR. Use the formula:

$$95\% \text{ CI} = \bar{x} \pm (2 \times \text{SE})$$

Key to symbols:

$\bar{x}$ = mean

SE = standard error

Figure 1.4 shows a graph of the results, with error bars showing the 95% CI.

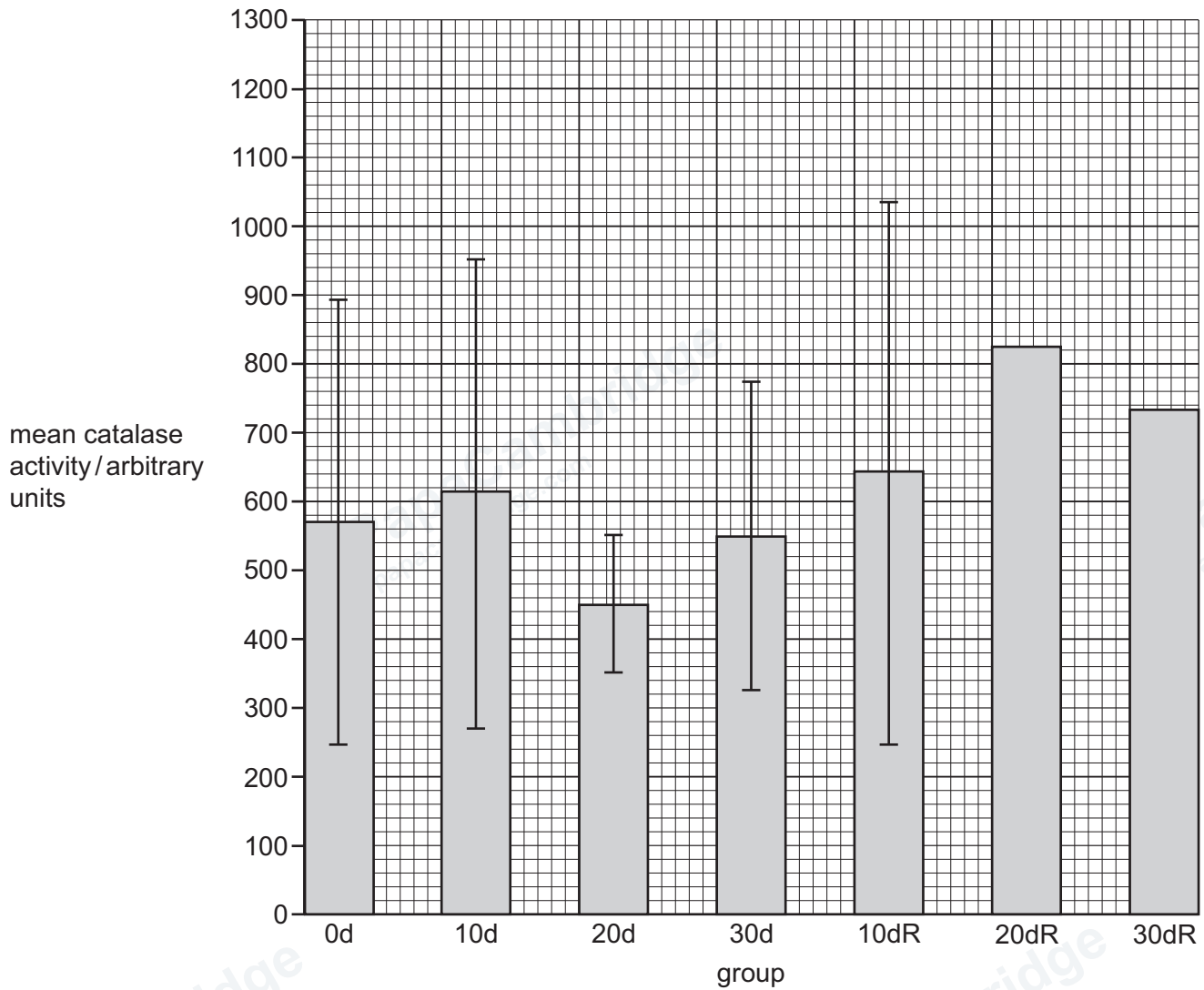
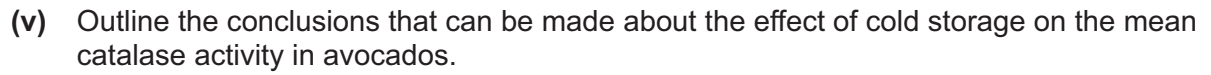


Figure 1.4

- (iv) On Figure 1.4, draw error bars showing the 95% CI for groups 20dR and 30dR that you calculated in 1(c)(iii). [2]



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- Suggest **two** reasons why the t -test is **not** an appropriate choice of statistical test for this investigation.

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[2]

[Total: 20]



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